

The Statistical Horizon: A Scaling Benchmark and Procedural Caution for Institutional Use of AI Text Watermarking

Henri F. Van Ryn
Independent Researcher
henri.vanryn@gmail.com

February 10, 2026 (revised June 23, 2026)
Version 5.8 — Preprint

Abstract

AI text watermarking has evolved from academic speculation into industrial practice. Yet watermark detections remain probabilistic indicators rather than proof of authorship. This paper formalizes the Statistical Horizon as a structural scaling benchmark for finite-sample watermark detectability. In the stylized mean-shift model used here, required sample length scales inversely with the square of the combined signal term $\eta\delta$, creating a steep, inverse-quadratic penalty for low-visibility watermarking.

The paper argues that this scaling result should not be confused with a rule for strong institutional attribution. Under the model developed here, the Statistical Horizon n_h is best understood as a lower detectability floor: it corresponds to the point at which expected detector signal reaches threshold and, in the Gaussian power analysis, to a 50% detection-power point. Stronger institutional use generally requires longer text than this floor alone would suggest. To make that gap explicit, the paper also defines a higher-power reliability benchmark n_E , showing that practical institutional interpretation depends not only on watermark strength but also on desired power, calibration, sample length, robustness, and disclosure. The resulting framework is intended to discipline how watermark signals are relied upon rather than to weaken watermarking technology as such.

Keywords: AI watermarking; statistical inference; forensic science; institutional reliance; machine learning governance; digital provenance; AI policy.

1 Introduction

1.1 Motivation and the Interpretive Gap

The rapid proliferation of Large Language Models (LLMs) has forced a parallel evolution in digital provenance. As synthetic text becomes increasingly difficult to distinguish from human writing at the semantic level, watermarking has emerged as a leading technical strategy for attribution. These approaches—ranging from synonymous substitution to logit “green-list” biasing—attempt to embed a recoverable statistical signal in generated text.

Yet the interpretive problem begins where the engineering result ends. Technical watermarking literature evaluates systems in terms of detectability: z -scores, likelihood ratios, power, and related performance measures. Institutional and adjudicative use, however, asks a different question: what, if anything, a detector output warrants as evidence. Those questions are related, but they are not

identical. A detector can be capable of signaling under some conditions without thereby yielding robust evidence of authorship in disciplinary, regulatory, or forensic settings.

Watermark detection is not the discovery of a hidden cryptographic tag. It is a probabilistic inference drawn from a finite text sample under model-dependent assumptions. This paper formalizes the Statistical Horizon: a structural scaling benchmark that marks the sample-length regime in which expected detector signal reaches a specified threshold under stated assumptions. The goal is not to undermine watermarking technology, but to discipline its evidentiary use. Watermark evidence should therefore be treated as an indicative and assumption-sensitive forensic signal, not as an oracle of authorship.

The paper’s central claim is narrow but consequential. The horizon equation identifies a structural relationship between watermark strength, detector response, and sample length. It does not by itself yield a universal admissibility threshold, nor does it license a fixed token rule across watermarking systems. On the contrary, the framework shows that required text length is highly sensitive to hidden or undisclosed parameters. In low-visibility regimes, the sample lengths needed for practically informative detection may be much larger than casual policy discussion assumes.

This matters because short-text detector outputs are especially vulnerable to overinterpretation. In settings where watermark strength, calibration data, and system parameters are not disclosed, external reviewers cannot reliably determine how much institutional weight a detector result deserves. The resulting problem is not merely technical but procedural: absent disclosure and corroboration, institutions risk treating a model-dependent signal as though it were dispositive evidence.

1.2 Related Work and Contribution

Kirchenbauer et al. show that watermark detection can succeed on very short samples under specific experimental conditions, including relatively strong watermark parameters such as $(\gamma, \delta) = (0.25, 2)$ in their proof-of-concept setting [1]. That result is important as an engineering demonstration of detectability, but it does not by itself establish a forensic or regulatory standard for interpreting detector outputs on short text. Subsequent work has emphasized robustness, distortion-free or unbiased watermarking, semantic or paraphrase robustness, public detectability, and attack surfaces such as stealing, spoofing, and scrubbing [2, 3, 4, 5, 6, 7]. Statistical work has also generalized the efficiency analysis of watermark detectors beyond simple z -tests [8].

Accordingly, this paper distinguishes three levels of analysis that are often blurred in discussion of AI watermarking: statistical detectability, practical reliability, and forensic or institutional use. Its contribution is not a new test statistic. Rather, it uses a simple scaling benchmark to show why low-visibility watermarking creates an unusually severe short-sample problem for institutional use, and why the detectability of a signal should not be confused with its reliability for institutional reliance.

That distinction has become more urgent as watermarking and provenance move from research demonstrations into live governance settings. In the European Union, transparency obligations under Article 50 of the AI Act become applicable on 2 August 2026,¹ and the Commission’s Code of Practice on Transparency of AI-Generated Content, published on 10 June 2026, now supports implementation of those Article 50 marking, detection, and labelling obligations on a voluntary basis. The present framework is not offered as an interpretation of any single legal instrument; rather, it is offered as a way to discipline evidentiary claims in a policy environment that increasingly expects marking, detectability, and transparent interpretation to coexist.

The interplay among the EU AI Act’s Article 50 transparency obligations, emerging state-level AI disclosure laws, and provenance standards such as C2PA increases the practical importance of

¹See Regulation (EU) 2024/1689, Art. 113 (application date); Art. 50 (transparency obligations) [16].

distinguishing compliance marks from reliable institutional evidence.² These legal and technical frameworks variously require, enable, or begin to standardize machine-readable indications of origin—at different scopes and with different legal force. But the existence of a mark, manifest, or provenance metadata is not the same as downstream reliability for institutional reliance. Compliance-oriented outputs should not be mistaken for decision-ready evidence.³ That observation applies most directly to probabilistic detection outputs; cryptographic provenance attestations such as C2PA manifests raise distinct verification questions outside the scope of this paper. What is needed—and what the present framework is intended to support—is disclosed reliability information showing when a detector output has been adequately characterized, in terms of error rates, text-length thresholds, and stated assumptions, for the institutional use to which it is being put.

2 The Statistical Horizon: Formalization

2.1 Key Assumptions

The horizon derivation in this paper rests on three stylized assumptions:

- **Approximate token-level independence:** Green-list assignments and token draws are treated as independent for analytic tractability.
- **Locally linear watermark response:** The increase in green-token probability is proportional to the logit bias δ for small-to-moderate δ .
- **Gaussian approximation:** The normalized detector statistic is approximated by a standard normal distribution under the null and a shifted normal under the alternative.

Each assumption may be violated in practice. Violations (especially positive autocorrelation) generally increase required sample length, making the stylized derivation optimistic from the perspective of detectability. The cautionary conclusion is therefore robust in the sense that many realistic departures from the stylized model do not make undisclosed short-sample attribution safer.

A rough sense of magnitude is useful. Under positive autocorrelation with lag-1 coefficient ρ , the variance of the green-token count exceeds the independence benchmark by an inflation factor that, for an AR(1)-like structure, is approximately $(1 + \rho)/(1 - \rho)$. Because the horizon scales inversely with the variance-normalized signal, the required sample length grows by roughly the same factor: a modest $\rho = 0.3$ implies an inflation of about 1.9 $\times$, and $\rho = 0.5$ implies roughly 3 $\times$. Token-level dependence is material because green-list assignment is itself seeded by preceding tokens, so corrections in this range are plausible rather than extreme. The independence-based horizon should therefore be read as a lower bound on required length, potentially by a factor of two or more at the text lengths institutions typically evaluate. This estimate is heuristic: the exact variance structure induced by green-list seeding and decoding is more complex than a simple binary AR(1), and the figures above are intended as order-of-magnitude intuition rather than calibrated corrections.

²For example, Article 50 of the EU AI Act addresses marking and detection of AI-generated content; California’s AI Transparency Act imposes disclosure and detection-tool obligations on covered generative-AI providers for specified categories of image, video, and audio content [17]; and C2PA provides a technical provenance framework through Content Credentials [14]. These regimes differ in legal force, scope, and technical mechanism.

³The phrase “decision-ready evidence” is used in a technical-policy sense; it does not denote a legal standard of admissibility or sufficiency under any evidentiary rule or jurisdiction-specific standard.

2.2 Derived Scaling Relation

In the simplified mean-shift model used here, the sample length at which the expected detector statistic reaches threshold under the normal approximation scales according to the relation derived in Appendix A. More generally, if $z^* > 0$ denotes the critical value of the detector statistic for a one-sided threshold test under the stylized normal approximation, and η denotes the variance-normalized detector-response parameter defined formally in Appendix A, the horizon length n_h satisfies:

$$n_h = \frac{(z^*)^2}{(\eta\delta)^2} \quad (1)$$

Definition 1 (Statistical Horizon). n_h is the sample length at which the expected detector statistic reaches the threshold z^* . Under the Gaussian model developed in Appendix B, this corresponds to a 50% detection-power point. It is a lower detectability floor, not a reliability benchmark for institutional reliance.

What this result does not mean. The Statistical Horizon n_h is not a universal admissibility threshold, a minimum token rule applicable across all watermarking systems, or a quantitative claim about any specific deployed detector. It is a structural scaling result derived under three stylized assumptions—approximate token independence, locally linear watermark response, and Gaussian approximation—that are stated precisely because they are known to be optimistic. The horizon marks the sample length at which expected detector signal reaches threshold under those assumptions; it does not mark the length at which a detector output becomes suitable for institutional reliance. That determination requires additional inputs: disclosed system parameters, empirically characterized error rates, robustness information, and a power analysis appropriate to the decision being made. The numerical examples in Table 2 are analytic sensitivity illustrations under the reduced-form model; they are not calibrated claims about real-world detector performance.

The scaling relation above is derived only under the formal definition of η . In the main text, references to “effective detector response” (η_e) are heuristic and intended for sensitivity reasoning under non-ideal conditions rather than exact cross-system calculation.

The operational value of n_h and n_E depends on whether η , or a deployment-specific effective analogue, can be disclosed, estimated, or bounded. The parameter η is not generally observable from a detector output alone; it depends on the watermark design, detector construction, and the model’s token-level response to the watermarking intervention. Where a provider does not disclose enough information to estimate or bound that response, the horizon cannot be externally computed. In that setting, the framework functions not as a usable threshold but as a disclosure requirement: the institution lacks the information needed to determine whether the detector output is reliable at the relevant text length.

Even when treated illustratively, the sensitivity of n_h to the combined signal is substantial because $n_h \propto 1/(\eta^2\delta^2)$. Halving the effective signal requires approximately four times as much text to maintain the same expected detector signal.

That distinction matters. Crossing a statistical detectability floor and satisfying a strong standard for institutional reliance are related but distinct events. The former concerns model-based signal detectability under specified assumptions. The latter concerns whether the resulting signal is sufficiently reliable, transparent, and interpretable to carry substantial institutional weight. That second judgment depends not only on the existence of a signal, but also on desired power, disclosure, calibration, robustness, and the availability of corroboration.

2.3 From a Detectability Floor to a Higher-Power Reliability Benchmark

If the target is not mere detectability but a higher operational power level $1 - \beta$, then the Gaussian mean-shift model yields a corresponding higher-power reliability benchmark:

$$n_E = \frac{(z^* + z_{1-\beta})^2}{(\eta\delta)^2} \quad (2)$$

where $z_{1-\beta} = \Phi^{-1}(1 - \beta)$ is the standard normal quantile associated with the desired power (e.g., $z_{0.8} \approx 0.84$, $z_{0.9} \approx 1.28$). Equation (2) makes explicit what the basic horizon alone can obscure: the amount of text needed for a higher-confidence institutional use may be several times larger than the 50%-power floor n_h . The notation n_E is retained for continuity with earlier drafts; it refers here to a higher-power reliability benchmark, not to legal evidentiary sufficiency. Throughout, “reliability” denotes the statistical power and repeatability of detection under the assumed model; it is distinct from the reliability standard of Federal Rule of Evidence 702(c)–(d), which Section 5 addresses separately.

For a fixed detectability threshold z^* , the gap between n_h and n_E is governed by the power term $z_{1-\beta}$. The ratio is:

$$\frac{n_E}{n_h} = \left(1 + \frac{z_{1-\beta}}{z^*}\right)^2 \quad (3)$$

For example, when $z^* = 3$, the required sample size increases by approximately $1.64\times$ for 80% power, $2.04\times$ for 90% power, and $2.40\times$ for 95% power. Thus the higher-power reliability benchmark does not merely ask whether detection has crossed a lower statistical floor; it asks how much additional sample length is required before an institution may rely on that detection with a specified level of power.

It is important to read n_E as a statistical quantity. It is the sample length required to achieve a chosen detection power against a specified alternative; it is not, by itself, a measure of institutional weight, which additionally depends on prior odds, the asymmetric costs of false positives and false negatives, robustness to model misspecification, and adversarial exposure. n_E is therefore best understood as a model-based power-analysis benchmark that can inform stronger institutional reliance, not a complete decision rule or standalone basis for attribution.

2.4 A Concrete Illustration of Overinterpretation

Consider a hypothetical academic-integrity setting. A 300-word student essay (≈ 400 tokens) yields a watermark detector output with $z = 4.2$, well above a typical threshold of $z^* = 3$. An administrator might treat this as strong evidence of AI generation. However, suppose the watermarking system operates in a low-visibility regime (e.g., $\eta\delta = 0.1$). Then $n_h = 9/0.01 = 900$ tokens, and n_E at 90% power is roughly $2.04 \times 900 \approx 1,836$ tokens. Crucially, n_h is a population-level benchmark on the expected detector signal, not a per-sample guarantee. A short text can produce a high observed statistic, and if the detector is well calibrated that statistic may still be evidence against the null. The institutional problem is different: at 400 tokens, the sample is below the length at which the assumed model would deliver the desired pre-specified detection power, and without disclosed parameters, comparable-length calibration, and dependence checks, the result should not be converted into strong attribution.

This example is illustrative, not a universal claim. But it shows why short samples and undisclosed parameters are a dangerous combination for institutional use. Even under these simplified assumptions, short-text attribution may remain institutionally underpowered after it has crossed a minimal detectability boundary.

2.5 The Horizon Is Operationally Dynamic

The Statistical Horizon should not be read as a static property of a watermarking problem once and for all. In practice, the effective signal term $\eta_e \delta$ can shift as watermark construction changes, detector design improves, post-generation editing alters the signal, or disclosure quality allows external reviewers to model the system more accurately. A fixed text length may therefore occupy different evidentiary positions at different times or across different deployments.

This point matters for governance. Institutions may be tempted to treat a detector output as though it had stable meaning across vendors, revisions, or attack conditions. The present framework counsels against that inference. The horizon is better understood as an operationally dynamic benchmark: a way to ask what sample length is required under stated conditions, not a way to assume that one detector result carries the same weight across changing technical or institutional settings.

3 Institutional Use and the “Safety Floor” Principle

The Statistical Horizon has direct implications for responsible institutional use of watermark evidence, but not in the form of a universal token threshold. The horizon equation identifies a scaling relationship between watermark strength, detector response, and sample length. It does not by itself establish that any fixed number of tokens is generally sufficient for forensic, disciplinary, or regulatory use across watermarking systems.

This limitation is not incidental. In real deployments, the quantities needed to interpret detector outputs may be partially or wholly undisclosed: watermark strength, green-list construction, detector design, calibration data, robustness to post-editing, and other system-specific parameters may not be available to external reviewers. Under those conditions, even a formally correct detector output may have uncertain evidentiary meaning.

AI watermark detection is therefore best understood as a hybrid evidentiary form: a machine-mediated statistical signal whose institutional weight depends on assumptions, calibration, sample length, and operator transparency. This cautionary stance is consistent with broader forensic-science literature, which has repeatedly stressed that scientific validity, calibrated limits, and transparent interpretation are prerequisites for responsible institutional use of technical evidence [9, 10]. The present paper’s narrower contribution is to show why watermark evidence is especially vulnerable to short-sample failure: low visibility and weakened effective response impose an inverse-quadratic sample-length penalty, making the gap between detectability and reliability for institutional reliance unusually severe.

3.1 The Safety Floor

In this paper, the Safety Floor refers to the procedural stance that treats n_h as a necessary detectability floor but not, standing alone, an adequate basis for attributing substantial institutional weight to a detector output, with stronger institutional claims requiring movement toward a higher-power benchmark such as n_E together with the additional inputs identified in Section 3.2. Where text is short, watermark strength is unknown, or calibration evidence is unavailable, the default institutional posture should be conservative.

Table 1 summarizes a proposed heuristic for weighing watermark signals under different conditions. This table is a diagnostic aid, not a decision procedure. Its rows do not create universal thresholds, admissibility rules, or categorical legal standards. They describe institutional postures under stated assumptions about detector calibration, text length, and parameter disclosure. Where

those parameters are estimated, undisclosed, or domain-dependent, the table should be read as uncertainty-aware guidance rather than a bright-line rule.

Table 1: Heuristic institutional reliance postures (one possible heuristic, illustrative rather than normative).

Condition	Suggested institutional posture
$L \ll n_h$ or parameters undisclosed	No standalone attribution; not externally weightable
$L \approx n_h$ but no calibration	Indicative only; requires corroboration
$L \geq n_E$ with disclosed parameters and robustness tests	May support moderate institutional weight
Non-watermark provenance available (e.g., C2PA)	Potentially stronger combined reliability, subject to verification

3.2 Disclosure and Corroboration

The framework supports a firm disclosure principle: if an institution wishes to rely materially on watermark evidence, it must disclose enough about the detector family, calibration regime, sample-length comparability, and robustness assumptions to allow an external reviewer to assess whether the sample meaningfully exceeds a disclosed detectability floor. In the absence of such disclosure, watermark evidence should be treated as unreliable for strong institutional use.

This principle is normative rather than formally derived from the horizon equation; it is proposed as a procedural response to parameter sensitivity, opacity, and finite-sample uncertainty. A minimally adequate showing in an academic-integrity setting might include disclosure of the detector family used, calibration results on texts of comparable length, reported false-positive behavior under ordinary editing or paraphrasing, and corroborating non-detector evidence tied to the same submission.

Watermark evidence should not be treated as existing in isolation from other possible attribution signals. A downstream institution may also consider stylometric indicators, linguistic structure, document organization, edit history, provenance metadata, or author-specific baselines where available. But those signals may not be independent. Many of these features may share a common source in the generative model’s underlying probability distribution, decoding process, or post-generation editing history, making their apparent independence an assumption rather than a demonstrated property. Structural correlations are plausible at multiple levels of the text and should not be assumed absent empirical demonstration: word choice, sentence construction, paragraph organization, document structure, argument style, and broader writing style. Treating each indicator as a separate and independent source of evidence can therefore overstate the strength of the overall attribution.⁴

The Statistical Horizon problem is one instance of a broader caution: assumptions about independence, error structure, and evidentiary strength must be examined rather than silently inherited. In high-stakes settings, an unexamined assumption can become the source of an incorrect conclusion.

⁴This is a structural inference rather than a claim about any specific detector or feature set. The empirical correlation structure among stylometric, organizational, provenance-based, and model-based indicators in AI-generated text has not been systematically characterized. Conditional independence should therefore be treated as a hypothesis requiring verification before multiple signals are combined as independent evidence.

3.3 Base Rates and Multiple Testing

The horizon analysis so far concerns a single text evaluated in isolation. Institutional deployment rarely looks like that, and two population-level effects further constrain the institutional weight of a positive detection.

First, base rates govern positive predictive value. Even a well-calibrated detector with a low false-positive rate α will, when applied to a large corpus that is mostly human-written, return a substantial number of false positives in absolute terms. The probability that a flagged text is in fact watermarked depends not only on α and detector power but on the prior prevalence of watermarked text in the population screened. Where that prevalence is low or unknown, a positive detection on a single short sample carries correspondingly weak posterior support, regardless of how far the observed statistic exceeds z^* . This is the familiar base-rate problem of forensic inference, and it compounds the short-sample fragility described above rather than substituting for it.

Second, multiple testing inflates expected false positives. An institution screening thousands of submissions performs thousands of simultaneous tests; a per-test threshold that appears stringent in isolation can produce many spurious flags across the corpus. Responsible institutional use therefore requires error control appropriate to the screening scale—for example, family-wise or false-discovery-rate adjustment—before any individual flag is treated as meaningful. Neither base-rate sensitivity nor multiple-testing correction is captured by n_h or n_E , which describe the detectability and power of a single test. They are additional constraints on responsible institutional reliance, not consequences of the horizon itself.

The higher-power reliability benchmark n_E is a power-analysis benchmark, not a posterior probability of AI generation or watermark presence. It indicates whether a detector has enough sample-length-supported signal to achieve a stated power level under the assumed model. Institutional decisions may also require base-rate information, likelihood ratios, alternative explanations, and assessment of asymmetric costs. Thus n_E can inform institutional reliance, but it does not by itself complete the Bayesian or decision-theoretic analysis. Even when a detector output meets or exceeds n_E under full disclosure and robustness conditions, institutional weight still requires base-rate adjustment, multiple-testing control where applicable, corroboration, and case-specific assessment of asymmetric costs; n_E remains an input to that judgment, not a substitute for it.

3.4 Adversarial Degradation

A further complication is adversarial degradation. In contemporary settings, the threat is not limited to benign paraphrasing or ordinary editing. Spoofing, watermark stealing or forgery, model adaptation, and cross-system attack strategies may either weaken a genuine signal or create a misleading appearance of one [7, 11, 12]. The structural intuition is that reductions in effective signal-to-noise ratio compound as a sample-length burden, while forged or counterfeited signals create a distinct false-positive risk. Both attack modes strengthen the case for cautious treatment unless robustness conditions are disclosed and empirically supported.

The cited results give a qualitative sense of magnitude. Watermark stealing and spoofing attacks can reconstruct enough of the green-list structure to forge or scrub signals, which in the framework’s terms drives effective $\eta\delta$ toward zero for scrubbing and creates false-positive signal where none was generated for spoofing [7, 11]. Adaptive stealing approaches reduce the text required to approximate the watermark rule, eroding the very disclosure-and-calibration assumptions on which reliable detection depends [12]. Analyses of production-oriented systems indicate that robustness varies substantially with editing and paraphrase intensity. The common structural point is that adversarial pressure does not merely add noise; it can shift the effective operating point of the detector in ways

that are not observable from the detector output alone, which is precisely the condition under which the framework counsels against strong institutional reliance.

3.5 Practical Summary

- A watermark “hit” on a short passage does not by itself establish origin.
- Reaching the detectability floor n_h does not by itself justify strong institutional weight.
- Where system parameters and robustness conditions are undisclosed, the default institutional posture should be cautious and ordinarily non-attributive.
- Stronger institutional use should require transparent calibration evidence, disclosure of relevant operating and robustness conditions, and corroboration from independent sources.

4 From Detection to Adjudication: A Four-Step Institutional Sequence

Watermark detection does not, by itself, determine the institutional status of an artifact. Even where a detector output supports an inference of synthetic involvement at a stated detection threshold and power level, institutions must still characterize the form and degree of human action surrounding the artifact’s production, revision, selection, verification, and use. A probabilistic indication of synthetic origin may be relevant to institutional consequence, but it is not enough by itself.

This paper accordingly distinguishes detection from adjudication through a four-step sequence organized in two phases. The sequence is schematic rather than jurisdiction-specific: different legal and policy regimes may attach different consequences to the same forensic signal. The same signal may do different institutional work: in court, it may bear on reliability; in regulation, it may trigger transparency obligations.

The technical phase comprises the paper’s quantitative contribution: (1) detection of possible synthetic involvement, and (2) statistical signal and power analysis relative to n_h and n_E ; it does not itself supply a legal or institutional decision rule. This phase asks whether the detector output supplies enough model-based signal, under stated assumptions, to justify considering institutional inquiry. Where a detector cannot satisfy these threshold requirements under the conditions of use, the sequence provides no basis for institutional escalation. It does not, and cannot, establish the institutionally relevant questions of human responsibility: what human action occurred, by whom, under what standard, and with what consequence.

The institutional phase comprises the work that must follow: (3) characterization of post-generative human action, meaning the professional, legal, or scholarly judgment that occurred after generation, and (4) analysis of institutional consequence, which depends on the specific duties and standards governing the relevant domain.

Step 3 marks the point at which this paper’s statistical analysis ends and institution-specific judgment begins. The relevant rubrics are domain-specific and lie outside mathematical verification. This paper does not propose a universal standard for characterizing post-generative human action; it identifies the question institutions must answer before a probabilistic signal is converted into consequence. This sequence is not a prediction that institutions will reliably preserve the distinction; it identifies the procedural boundary they must preserve if probabilistic detection is to support legitimate institutional consequence.

Consider, as one concrete illustration from legal practice, an AI-assisted legal brief. The detector output may satisfy the paper’s higher-power benchmark n_E under the assumed model.

The relevant institutional question, however, is not whether AI participated in generation. It is whether the attorney verified the citations and legal propositions, exercised independent professional judgment, and satisfied applicable procedural and professional duties, including Rule 11 certification, competence, candor, and supervision. A detector may support an inference that AI systems contributed to the artifact’s production; it cannot, alone, establish whether the attorney’s post-generative action met or failed those existing duties. The institutional consequence of detection therefore depends not on the technical finding in isolation, but on the application of established professional and procedural standards to the artifact’s full production history.

Characterizing post-generative human action requires evidence categories that institutions can actually demand and examine. Depending on the setting, these may include prompt logs showing what was requested of the system; timestamped edit history with authorship attribution; version diffs documenting how the artifact changed after generation; human verification traces, such as records of citation-checking or fact-checking; and contemporaneous statements of the role the author understood the tool to play. None of these is dispositive alone, and their availability varies by domain. But specifying the categories converts Step 3 from a procedural gesture into a checklist an institution can be held to, which is the point: the boundary between detection and adjudication is only meaningful if the adjudication side has identifiable inputs.

This distinction reinforces the broader claim of this paper: detection is not adjudication. Provenance signals may contribute evidentiary context, but they cannot independently resolve authorship, professional responsibility, or institutional consequence. Institutional frameworks that treat synthetic detection as self-interpreting risk collapsing materially distinct categories of post-generative human action into a single procedural label.

Synthetic origin may answer *how* an artifact was produced; it does not determine *what* the artifact is evidence of, nor what institutional response detection licenses.

5 Limitations and Scope

This paper is a structural and procedural argument, not a technical or empirical contribution, and several limitations follow directly from that scope.

First, the framework introduces no new detector, test statistic, or empirical evaluation. The scaling relation is a consequence of a standard signal-detection model, and the numerical values in Table 2 are analytic illustrations rather than measurements calibrated to any deployed system. Applying the framework to a specific watermarking implementation would require estimating η and δ for that system, which this paper does not attempt.

Second, the derivation rests on stylized assumptions: token-level independence, locally linear watermark response, and a Gaussian approximation. Realistic departures, especially positive autocorrelation and non-stationarity, generally increase the required sample length, so the formulas are optimistic from the standpoint of detectability. The paper establishes the direction of this correction but not its magnitude; quantifying it under realistic mixing conditions is left to future work.

Third, n_h and n_E are statistical quantities describing the detectability and power of a single test. They do not encode prior odds, the asymmetric costs of false positives and false negatives, robustness to model misspecification, or adversarial tampering. As Section 3 emphasizes, base rates, multiple-testing control, disclosure, and corroboration are additional inputs or constraints for institutional reliance; the benchmarks are inputs to that judgment, not substitutes for it.

Fourth, the legal and policy discussion is deliberately schematic. The four-step sequence identifies a procedural boundary; it does not map the proposed benchmarks onto specific standards of proof

(for example, preponderance of the evidence versus beyond a reasonable doubt), nor does it work through a particular adjudicated case. Those mappings are domain-specific and lie outside the statistical scope of this paper.

This paper avoids using “sufficiency” as a label for its proposed statistical benchmarks. In evidence law, “sufficiency,” “conditional relevance,” “admissibility,” and “expert reliability” have specific meanings under rules such as Rule 104(b), Rule 702, Daubert, and related state-law standards: Rule 104(b) concerns whether evidence is sufficient to support a finding of a conditional fact, and the 2023 amendment to Rule 702 requires the proponent to establish, more likely than not, that expert testimony satisfies each of the rule’s requirements. The Statistical Horizon framework instead uses the language of detectability, reliability, power, and institutional reliance. Its proposed benchmarks ask whether a detector output has enough sample-length-supported signal to inform a proposed attribution inference under stated assumptions; they do not determine admissibility, legal sufficiency, or ultimate institutional responsibility. The paper accordingly does not propose a substitute for Rule 104(b), Rule 702, Daubert, any state-law analogue such as Frye or hybrid admissibility standards, or any jurisdiction-specific evidentiary rule, nor does it purport to supply a governing standard for non-judicial institutional contexts not governed by the Federal Rules of Evidence.

If a watermark detector output were offered through expert testimony in a federal proceeding, Rule 702 would require the proponent to establish that the testimony rests on sufficient facts or data, reliable principles and methods, and reliable application to the facts of the case. As clarified by the 2023 amendment, this is a Rule 104(a) admissibility determination, made by the court on a more-likely-than-not standard, rather than a Rule 104(b) question of conditional relevance. The disclosures proposed here—text-length reliability curves, stated false-positive and false-negative rates, robustness assumptions, and limits on use—are technical inputs that may inform a court’s gatekeeping analysis, but they do not determine admissibility or satisfy Rule 702 on their own.⁵ The framework likewise does not address Rule 401 relevance, authentication, hearsay, or other ordinary evidentiary doctrines. A detector output might be minimally relevant in the Rule 401 sense while still failing Rule 702 reliability, Rule 104(a) gatekeeping, institutional policy requirements, or a merits standard of proof.

These limitations do not undermine the central claim; they locate it. The contribution is to discipline how a probabilistic signal is interpreted, not to certify any particular detector or to resolve any particular institutional case.

6 Conclusion: The Structural Limit of Probabilistic Attribution

AI text watermarking represents a meaningful advance in digital provenance. Yet its significance for institutional reliance is constrained not only by implementation choices, but by structural limits imposed by statistical inference on finite text samples. This paper formalizes the Statistical Horizon as a scaling benchmark for that problem and argues that detectability should not be confused with reliability for institutional reliance.

The resulting claim is narrower than a universal admissibility rule, but more robust. As watermark strength decreases and operating conditions become less transparent, required sample length rises sharply and interpretation becomes increasingly dependent on hidden parameters, detector calibration, and disclosure. Under such conditions, short-text detections are especially poor candidates for strong institutional attribution.

⁵See Fed. R. Evid. 702; *Daubert v. Merrell Dow Pharm., Inc.*, 509 U.S. 579 (1993); *Frye v. United States*, 293 F. 1013 (D.C. Cir. 1923).

The paper’s contribution is not a new statistical test. It is the use of a simple scaling benchmark to show why low-visibility watermarking creates an unusually severe short-sample problem for institutional use, and why a detectability floor may sit well below a plausible reliability benchmark. In that sense, the framework can inform disclosure and reliance weighting for probabilistic watermark signals, but it does not itself determine admissibility, proof, or institutional outcomes.

In an era of layered provenance and probabilistic attribution, structural limits are not obstacles to governance. They are conditions for governing responsibly.

Acknowledgments

The author thanks George Paras for technical review and for helpful signal-processing perspective that informed the discussion of correlation, independence, and effective sample-size reasoning in Section 2.1 and Appendix A.4.

A Derivation of the Statistical Horizon

The Statistical Horizon marks the model-based point at which the expected detector statistic reaches a specified threshold in a finite text sample. The derivation starts from the canonical green-list logit-biasing framework of Kirchenbauer et al. [1] and abstracts to a variance-normalized detector-response parameter η .

A.1 Preliminaries: Watermarking Generative Process

At each generation step t , the LLM produces logits $\ell^{(t)} \in \mathbb{R}^{|V|}$. A secret pseudorandom function (seeded by preceding tokens) partitions the vocabulary into a green list G_t of relative size $\gamma \in (0, 1)$. The watermarking rule adds a positive logit bias $\delta > 0$ to green-list tokens. After softmax normalization, the probability of sampling a green token rises from a baseline value p_0 to a perturbed value $p_1(\delta)$. For small-to-moderate δ , the induced probability lift may be linearized: $p_1(\delta) \approx p_0 + \kappa\delta$, where κ is a local response coefficient determined by the underlying token distribution and green-list construction. Formally, $\kappa = dp_1/d\delta|_{\delta=0}$; in the idealized case where baseline green-token mass is approximately γ , the local response is on the order of $\gamma(1 - \gamma)$, so that smaller green-list fractions and weaker biasing both reduce κ and thereby enlarge the required sample length through Equation (1).

A.2 Classical z -Test Formulation

Suppose a detector observes L generated tokens and counts the number of green tokens $X = \sum_t \mathbf{1}\{w_t \in G_t\}$. Under H_0 (no watermark): $E[X] = Lp_0$, $\text{Var}(X) \approx Lp_0(1 - p_0)$. The normalized detector statistic is:

$$Z = \frac{X - Lp_0}{\sqrt{Lp_0(1 - p_0)}} \quad (4)$$

Under H_1 (watermark): $E[X] \approx L(p_0 + \kappa\delta)$. Define $\eta := \kappa/\sqrt{p_0(1 - p_0)}$. Then $\mu_g = \eta\delta\sqrt{L}$.

A.3 Horizon Length

The Statistical Horizon is defined as the sample length n_h at which the expected detector statistic under H_1 reaches z^* : $\eta\delta\sqrt{n_h} = z^*$. Solving:

$$n_h = \frac{(z^*)^2}{(\eta\delta)^2} \quad (5)$$

A.4 Interpretive Limits

The derivation assumes a z -test-like detector, approximate token independence, and a locally linear watermark response. A practically important point is the effect of dependence. Because green-list assignment is seeded by preceding tokens and generated text is itself autocorrelated, exact token-level independence is not expected in realistic samples. Under positive token-level correlation, the variance of X exceeds the independence approximation, so the formula above tends to understate required sample length. In that sense, the stylized derivation is plausibly optimistic: realistic dependence would typically strengthen, not weaken, the paper’s cautionary conclusion.

B Power Analysis and the Higher-Power Reliability Benchmark

Under the normal-approximation model, $Z \mid H_1 \sim \mathcal{N}(\mu, 1)$ where $\mu = \eta\delta\sqrt{L}$. For a one-sided threshold test rejecting when $Z \geq z^*$, power is:

$$\text{Power}(L) = 1 - \Phi(z^* - \eta\delta\sqrt{L}) \quad (6)$$

At $L = n_h$, $\mu = z^*$, so $\text{Power}(n_h) = 1 - \Phi(0) = 0.5$. Thus the Statistical Horizon corresponds to a 50% detection-power point.

If the target power is $1 - \beta$, one must solve $1 - \Phi(z^* - \eta\delta\sqrt{L}) = 1 - \beta$, which yields $\eta\delta\sqrt{L} = z^* + z_{1-\beta}$. Hence the higher-power reliability benchmark is given by Equation (2).

For illustration, let $z^* = 3$. Then $n_h = 9/(\eta^2\delta^2)$. Table 2 gives illustrative values for n_h and n_E at 80% and 90% power.

Table 2: Illustrative detectability floors and higher-power reliability benchmarks ($z^* = 3$). Analytic sensitivity examples only; not calibrated claims about any specific detector or deployment.

η	δ	$\eta\delta$	n_h	n_E (80%)	n_E (90%)
0.8	0.8	0.64	22.0	36.0	44.8
0.8	0.4	0.32	87.9	144.1	179.0
0.8	0.2	0.16	351.6	576.5	716.1
0.4	0.4	0.16	351.6	576.5	716.1
0.4	0.2	0.08	1406.2	2305.9	2864.3
0.2	0.2	0.04	5625.0	9223.8	11457.3

C Extension Notes on Detector Classes

The formal derivation in Appendix A is specific to a z -test-style mean-shift detector. Other detector families, including likelihood-ratio or Bayesian detectors, may have different operating constants,

calibration procedures, and robustness profiles. The present paper does not claim that all such detectors can be reduced exactly to the same one-parameter representation.

What the paper claims is narrower. Across detector families, finite-sample performance plausibly depends on the magnitude of the induced separation between null and alternative distributions and on the amount of text available. The policy argument of the main text therefore does not rest on the claim that all detectors behave identically. It rests on the more modest point that opaque, parameter-sensitive, finite-sample statistical methods should not be given strong institutional weight without transparent support.

References

- [1] Kirchenbauer, J., Geiping, J., Wen, Y., Katz, J., Miers, I., and Goldstein, T. (2023a). A watermark for large language models. In *Proceedings of the 40th International Conference on Machine Learning*.
- [2] Kirchenbauer, J., et al. (2024). On the reliability of watermarks for large language models. *International Conference on Learning Representations*.
- [3] Kuditipudi, R., Thickstun, J., Hashimoto, T., and Liang, P. (2024). Robust distortion-free watermarking for language models. *Transactions on Machine Learning Research*.
- [4] Hu, Z., Chen, L., Wu, X., Wu, Y., Zhang, H., and Huang, H. (2024). Unbiased watermark for large language models. *ICLR*.
- [5] Ren, J., et al. (2024). A robust semantics-based watermark for large language models against paraphrasing. *Findings of ACL: NAACL 2024*, pp. 613–625.
- [6] Fairuze, J., et al. (2023). Publicly-detectable watermarking for language models. arXiv:2310.18491.
- [7] Jovanović, N., Staab, R., and Vechev, M. (2024). Watermark stealing in large language models. *ICML*.
- [8] Li, X., Ruan, F., Wang, H., Long, Q., and Su, W.J. (2024). A statistical framework of watermarks for large language models. arXiv:2404.01245.
- [9] National Academy of Sciences. (2009). *Strengthening Forensic Science in the United States: A Path Forward*. National Academies Press.
- [10] PCAST. (2016). *Forensic Science in Criminal Courts: Ensuring Scientific Validity of Feature-Comparison Methods*. Executive Office of the President.
- [11] Omid, R., Dong, Y., and Wang, B. (2026). On Google’s SynthID-Text LLM Watermarking System: Theoretical Analysis and Empirical Validation. arXiv:2603.03410.
- [12] Zhang, S., Chen, Y., Han, J., Cheng, B., and Ma, J. (2026). Beyond a fixed seal: Adaptive stealing watermark in large language models. arXiv:2604.10893.
- [13] Christ, M., Gunn, S., and Zamir, O. (2023). Undetectable watermarks for language models. arXiv:2306.09194.

- [14] Coalition for Content Provenance and Authenticity (C2PA). (n.d.). C2PA: Verifying media content sources. Accessed June 23, 2026. <https://c2pa.org/>
- [15] European Commission. (2026). Code of Practice on Transparency of AI-Generated Content. Published 10 June 2026. <https://digital-strategy.ec.europa.eu/en/policies/code-practice-ai-generated-content>
- [16] European Parliament and Council. (2024). Regulation (EU) 2024/1689 of 13 June 2024 laying down harmonised rules on artificial intelligence (Artificial Intelligence Act). *Official Journal of the European Union*, OJ L, 2024/1689, 12 July 2024.
- [17] California Legislature. (2024). SB-942, California AI Transparency Act, Chapter 25, California Business and Professions Code §22757 et seq. Operative January 1, 2026.